

The two-matrix formula for a periodic tridiagonal numerical range

Literature answer to Conjecture 3.7 in arXiv:1910.00720

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Status: literature-already-answered, full conjecture. For every n , the closure of the numerical range of the periodic tridiagonal operator in Conjecture 3.7 is the convex hull of the numerical ranges of the two stated $(n + 1) \times (n + 1)$ matrices. The later paper arXiv:2103.01866 explicitly proves this conjecture.

Source conjecture

Let b be the $(n + 1)$ -periodic sequence with period word $0^n 1$, and let $T = T(b, 0, 1)$ be the associated tridiagonal operator. Let B_n be the $(n + 1) \times (n + 1)$ matrix with ones on the first superdiagonal and zeros elsewhere, and let

$$J_n = \text{diag}(1, 0, \dots, 0, 1).$$

Conjecture 3.7 of Itzá-Ortiz–Martínez-Avendaño, on source PDF page 19, asserts that

$$\overline{W(T)} = \text{co}(W(B_n + J_n) \cup W(B_n - J_n)). \quad (1)$$

The source proves the case $n = 1$ and reports numerical verification for $n = 2, 3$, but says that it cannot yet provide a proof [1]. Its introduction also records that the referee had proposed a proof expected to appear later.

Explicit later solution

The abstract of Itzá-Ortiz–Martínez-Avendaño–Nakazato states that the paper proves a conjecture of the first two authors, and its introduction identifies that conjecture as Conjecture 3.7 of the source [2].

Their Theorem 2.3, beginning on supporting PDF page 7, proves a more general statement. For suitable real $(n + 1)$ -periodic lower- and upper-diagonal words a, c satisfying a palindromic system of modulus equalities, it constructs two explicit tridiagonal matrices A^+ and A^- and proves

$$\overline{W(T(a, 0, c))} = \text{co}(W(A^+) \cup W(A^-)).$$

Example 2.4, on supporting PDF pages 11–12, takes a with period word $0100 \cdots 0$ and c with period word $11 \cdots 1$. The hypotheses of Theorem 2.3 hold, and its two matrices are exactly $B_n + J_n$ and $B_n - J_n$. After passing by unitary equivalence to the period word $0^n 1$ used in the source, the example concludes explicitly that it proves Conjecture 3.7. Consequently, (1) holds for every n .

Mechanism and scope

The proof encodes every support function of the infinite operator’s numerical range in a homogeneous Kippenhahn polynomial. Under the palindromic hypotheses, this polynomial factors into the Kippenhahn polynomials of A^+ and A^- . The block-diagonal matrix $A^+ \oplus A^-$ therefore has the same closed numerical range, and its numerical range is the convex hull of the two block numerical ranges.

The later result is stronger than the source conjecture because it handles a larger palindromic family. The exact specialization leaves no part of Conjecture 3.7 open. This packet records an explicit literature answer, not an agent-generated proof.

Evidence checked

The source and supporting arXiv sources were rebuilt locally. The decisive locations are source PDF page 19, supporting Theorem 2.3 on page 7, and supporting Example 2.4 on pages 11–12. The arXiv records and journal DOI metadata were also checked.

References

- [1] B. A. Itzá-Ortiz and R. A. Martínez-Avendaño, *The numerical range of a class of periodic tridiagonal operators*, arXiv:1910.00720; Linear Multilinear Algebra **69** (2021), 786–806, doi:10.1080/03081087.2019.1706438.
- [2] B. A. Itzá-Ortiz, R. A. Martínez-Avendaño, and H. Nakazato, *The numerical range of some periodic tridiagonal operators is the convex hull of the numerical ranges of two finite matrices*, arXiv:2103.01866; Linear Multilinear Algebra **69** (2021), 2830–2849, doi:10.1080/03081087.2021.1957760.